MINI PROJECT

EduPredict – Student Academic Performance Prediction System

Github url: https://github.com/sri2-star/BDML-Mini-Project

1. Aim

The aim of this project is to develop a web-based machine learning application that predicts a student's **Performance Index (PI)** on a scale of **10 to 100** by analyzing various academic and lifestyle factors such as study hours, previous scores, sleep duration, attendance, extracurricular activities, and exam preparation. The system provides accurate predictions along with personalized recommendations to help students improve their academic performance.

2. Objectives

- To collect and preprocess student academic and lifestyle data.
- To develop a machine learning model capable of accurately predicting the student's Performance Index.
- To design a user-friendly web interface for entering student details.
- To generate personalized suggestions based on the predicted performance.
- To help students identify strengths and areas that require improvement.
- To support educators in monitoring student performance and providing timely guidance.
- To improve academic planning through data-driven decision making.

Tools and Technologies Used

Frontend

- HTML5
- CSS3
- JavaScript
- Bootstrap

Backend

- Python

- Flask

Machine Learning

- Scikit-learn
- Pandas
- NumPy

Data Visualization

- Matplotlib
- Seaborn

Development Environment

- Visual Studio Code

Dataset

- Student Performance Dataset (CSV)

Version Control

- Git
- GitHub

Deployment (Optional)

- Render / Railway / Heroku

Machine Learning Model

The project uses supervised machine learning algorithms to predict the Performance Index.

Possible algorithms include:

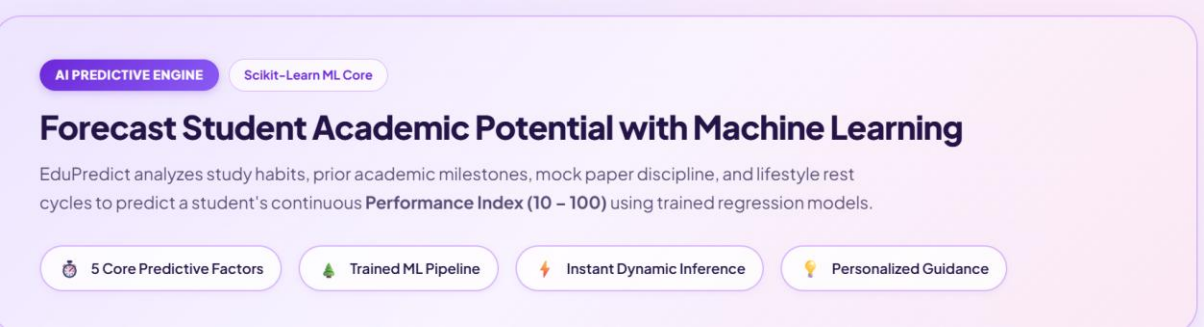
- Linear Regression
- Random Forest Regressor
- Decision Tree Regressor
- Gradient Boosting Regressor

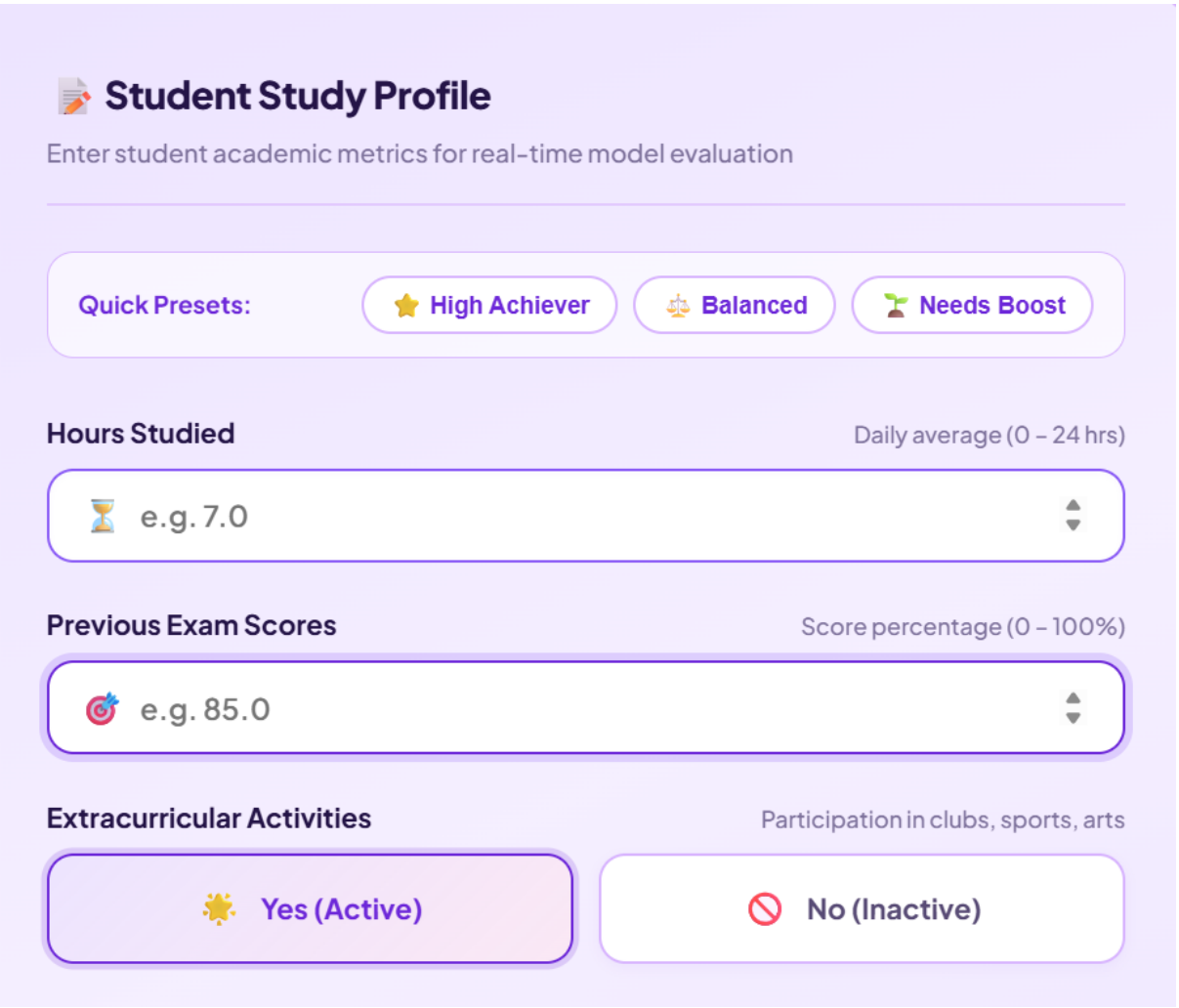
Among these, the model with the highest prediction accuracy is selected after evaluation using metrics such as:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)

- Root Mean Squared Error (RMSE)
- R^2 Score

OUTPUT SCREENSHOOTS:





Previous Exam Scores

Score percentage (0 – 100%)

🎯 e.g. 85.0

Extracurricular Activities

Participation in clubs, sports, arts

🌟 Yes (Active)

🚫 No (Inactive)

Sleep Hours

Daily average rest (0 – 24 hrs)

🌙 7.5

Sample Question Papers Practiced

Total mock tests solved

📑 4

Predict Performance Index

Reset

EduPredict

Student Academic Performance Analytics

Quick Presets: ⭐ High Achiever | ⚖️ Balanced | 🏋️ Needs Boost

Hours Studied — Daily average (0 – 24 hrs)

⏳ 2.5

Previous Exam Scores — Score percentage (0 – 100%)

🎯 48

Extracurricular Activities — Participation in clubs, sports, arts

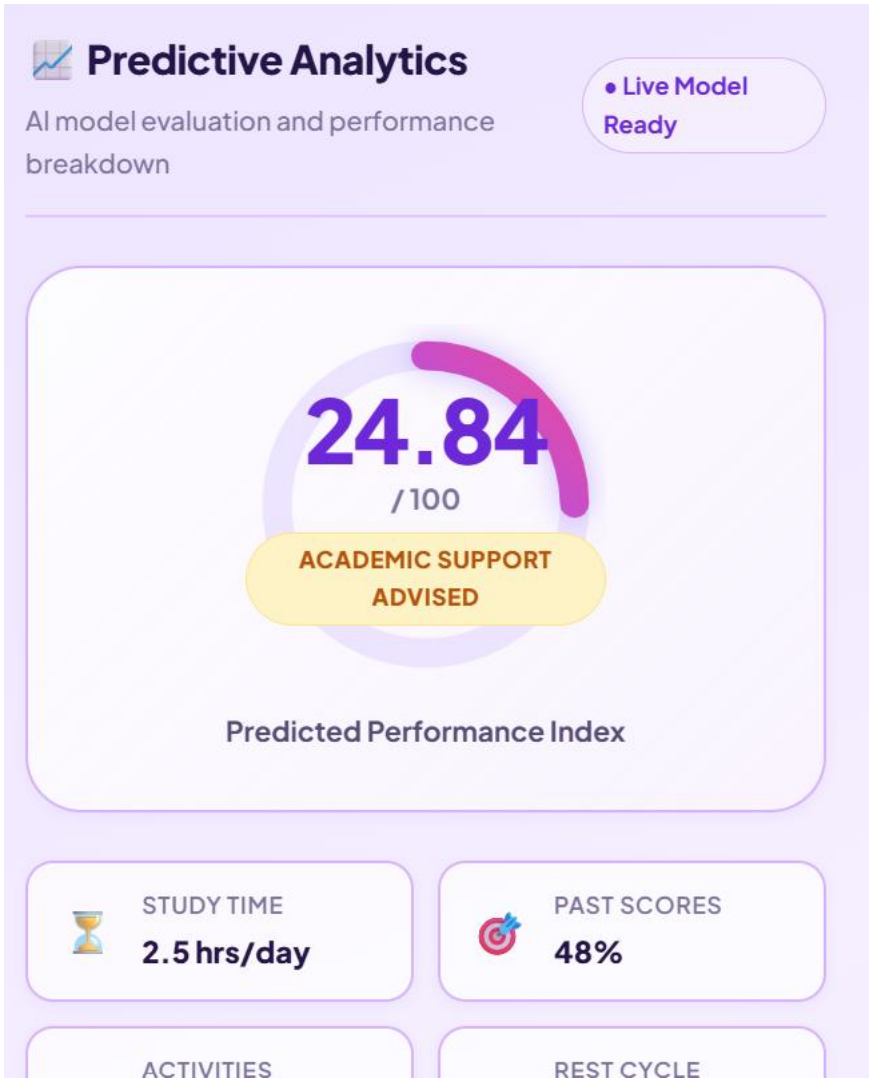
🌟 Yes (Active) | 🚫 No (Inactive)

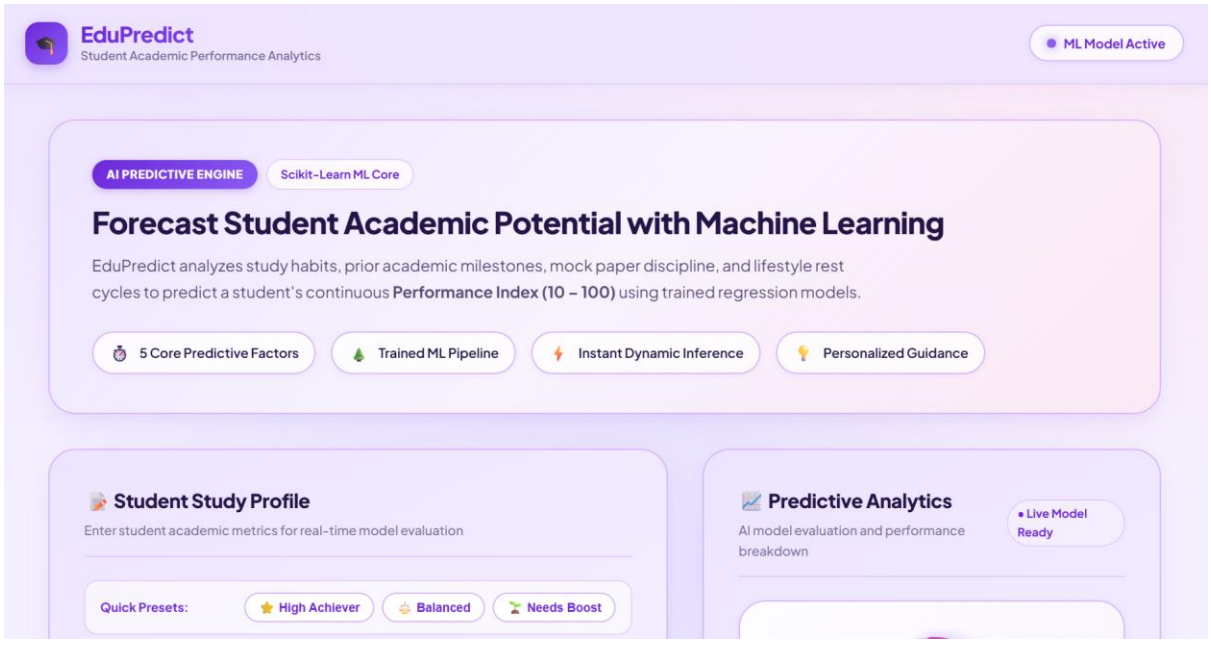
Sleep Hours — Daily average rest (0 – 24 hrs)

🌙 5.5

Sample Question Papers Practiced — Total mock tests solved

📑 1


Predictive Analytics
AI model evaluation and performance breakdown
• Live Model Ready
24.84
/ 100
ACADEMIC SUPPORT ADVISED
Predicted Performance Index
STUDY TIME
2.5 hrs/day
PAST SCORES
48%
ACTIVITIES
REST CYCLE


EduPredict
Student Academic Performance Analytics
ML Model Active
AI PREDICTIVE ENGINE
Scikit-Learn ML Core
Forecast Student Academic Potential with Machine Learning
EduPredict analyzes study habits, prior academic milestones, mock paper discipline, and lifestyle rest cycles to predict a student's continuous Performance Index (10 – 100) using trained regression models.
5 Core Predictive Factors
Trained ML Pipeline
Instant Dynamic Inference
Personalized Guidance
Student Study Profile
Enter student academic metrics for real-time model evaluation
Quick Presets:
High Achiever
Balanced
Needs Boost
Predictive Analytics
AI model evaluation and performance breakdown
• Live Model Ready